

EC3214 — International Finance

Summer 2023 Examination · Paper 1 · Full Marking Scheme

Module	EC3214 — International Finance
Internal Examiner	Ms. Aisling Finucane
Institution	University College Cork
Session	Summer 2023, Paper 1
Duration	1.5 hours
Structure	Section A — Long Questions: attempt 2 of 3 (100% of the paper). All questions carry equal weight.
Materials	Non-programmable calculator permitted

Coverage note. This scheme answers **all three** Section A questions in full, since you'll choose your best two under exam conditions.

Consistency with other sittings, and one important contrast. A1(a)'s flexible-price monetary model assumptions are word-for-word consistent with the EC3214 Summer 2024 scheme. A1(b) asks about the SAME four types of shock as that scheme's Q. A2(b) — but under a **fixed** exchange rate regime rather than a floating one, which completely changes which variable does the adjusting. Read the regime specified in each question carefully — the two scenarios have genuinely different correct answers, not just different wording.

Own-figure rule. A2(b), A2(c), and A3(b) all involve multi-step numerical calculations — an answer that applies the correct method consistently from an earlier arithmetic slip still earns the marks available for the later steps.

Source basis. A1(b)'s fixed-exchange-rate analysis follows the standard monetary

approach to the balance of payments (Frenkel and Johnson), and A3(a)'s discussion of the "UIRP puzzle" follows the standard treatment (Fama, 1984) of the forward premium puzzle — both flagged since they weren't set out in full in the uploaded course material.

Section A · Question 1

The flexible-price monetary model under a fixed exchange rate regime — 100 marks

A1(a)

[40 marks]

Explain the assumptions underlying the flexible price monetary approach model.

A1(b)

[60 marks]

Explain using the flexible price monetary approach model how each of the following will affect the home country exchange rate under a **fixed exchange rate regime**: (i) An expansion in the home country money supply. (ii) A foreign price increase. (iii) A rise in real income in the domestic economy.

FULL MODEL ANSWER

- **Purchasing power parity holds continuously:** absolute PPP ($S = P/P^*$) is assumed to hold at every point in time, not just in the long run — the exchange rate is always exactly consistent with relative price levels.
- **Prices are fully flexible:** both domestic and foreign price levels adjust instantaneously to clear their respective money markets, with no stickiness of any kind.
- **Output is fixed at its full-employment level:** real income y is determined independently of monetary factors (by the economy's real productive capacity), so monetary shocks cannot move output — **money is neutral**.
- **Stable money demand function:** real money demand is a stable, well-behaved function of income (and, in extended versions, the interest rate), so that $M_0^s = kPy$ (or its interest-rate-augmented version) reliably describes money market equilibrium.
- **Perfect capital mobility / uncovered interest parity:** financial markets are assumed to be perfectly integrated internationally, so that interest rate differentials reflect expected exchange rate movements.

Mark Allocation — Part (a) (40 marks)

Tier	Marks	What separates this tier
31-40	31-40	At least four of the five core assumptions correctly stated and explained

(continuous PPP, full price flexibility, fixed/full-employment output, stable money demand, perfect capital mobility/UIP).

21-30	21-30	At least three assumptions correctly stated.
11-20	11-20	Only one or two assumptions correctly stated, or stated without explanation.
0-10	0-10	No credible assumptions given.

The key structural difference from the floating-rate case: under a **fixed** exchange rate, S is pinned by policy — it is no longer free to adjust at all. Since absolute PPP still holds ($P = \bar{S}P^*$) and S is now fixed, the domestic price level P is entirely determined by the exogenous foreign price level P^* : $P = \bar{S}P^*$, where $\bar{S}$ denotes the fixed rate. This is the central insight of the monetary approach to the balance of payments (MABP): under fixed rates, the country **imports** its price level from abroad via the peg, and the variable that does the adjusting to keep the money market in equilibrium is no longer S — it is the **domestic money supply itself**, which adjusts endogenously through **balance-of-payments flows** (the central bank gaining or losing foreign reserves) rather than through any deliberate domestic monetary policy choice.

Shock	What happens to P	What happens to S	What happens to M_o^s (via reserves)
(i) Home money supply expansion	Unchanged (still pinned at $\bar{S}P^*$)	Unchanged (fixed by policy)	Falls back to original level — BoP deficit drains reserves, fully offsetting the expansion
(ii) Foreign price increase ($P^* \uparrow$)	Rises ($P = \bar{S}P^*$, $\bar{S}$ fixed, P^* up)	Unchanged (fixed by policy)	Rises — BoP surplus, reserves flow in to accommodate the higher money demand at the new, higher P
(iii) Domestic real income rise ($y \uparrow$)	Unchanged (still pinned at $\bar{S}P^*$)	Unchanged (fixed by policy)	Rises — BoP surplus, reserves flow in to accommodate higher money demand ($ky \uparrow$) at the unchanged price level

(i) Home money supply expansion — the loss of monetary independence: if the central bank tries to expand the domestic money supply independently (e.g. via domestic credit creation), the resulting money supply exceeds what money demand requires at the price level pinned by the peg ($P = \bar{S}P^*$, unchanged). The excess money spills into higher demand for foreign goods and assets, causing a **balance-of-payments deficit** — the central bank must sell foreign reserves to defend the fixed rate, which mechanically **contracts the money supply back down** to its original level. The net result: the exchange rate stays fixed (by definition), and the money supply itself ends up unchanged too — the attempted expansion is entirely reversed by the required reserve outflow. This is the monetary-approach mirror image of the Mundell-Fleming result that monetary policy is ineffective under fixed rates with high capital mobility (see the EC3213 Winter 2025 scheme, Q. A2(c)).

(ii) Foreign price increase — imported inflation: with $\bar{S}$ fixed and P^* rising, PPP forces P to rise proportionally — domestic prices are effectively **imported** from abroad via the peg. This higher P raises real money demand (kPy), and since the money supply hasn't independently changed, the resulting excess money demand is satisfied by a **balance-of-payments surplus** — reserves flow in, and the money supply **rises** endogenously to accommodate the higher money demand at the new, higher price level. The exchange rate

itself never moves (it can't, by construction); the entire adjustment happens through domestic prices and reserves instead.

(iii) Domestic real income rise — a pure reserve inflow, with no price change at all: a rise in y raises money demand ($k_y \uparrow$) directly, with no corresponding change in P (which stays pinned at $\bar{S}P^*$, unaffected by a purely domestic real shock). Since the money supply hasn't changed independently either, there is an excess demand for money at the existing price level — again resolved via a **balance-of-payments surplus**, with reserves flowing in and the money supply rising to meet the higher money demand. Note the contrast with the floating-rate case (see the EC3214 Summer 2024 scheme, Q. A2(b)): under floating rates, a rise in y causes *deflation* (P falls) and currency *appreciation*; under fixed rates, P cannot move at all, and the entire adjustment instead falls on the money supply via reserve inflows.

EXAM TECHNIQUE

The single clearest way to organise this answer is the summary table above: under fixed rates, S is *always* unchanged by construction (that's what "fixed" means) — the real content of each answer is in explaining which of P or M_o^s (via reserves) does the adjusting instead, and why. A student who answers this question by describing S moving in each case has misread the regime specified in the question and should be redirected to the floating-rate mechanics used in the EC3214 Summer 2024 scheme instead.

Mark Allocation — Part (b) (60 marks)

Tier	Marks	What separates this tier
46-60	46-60	Correctly establishes that S cannot move under a fixed regime, and that P is pinned at $\bar{S}P^*$ AND all three shocks correctly analysed in terms of which variable (P or M_o^s via reserves) adjusts and in which direction AND the loss-of-monetary-independence result for shock (i) is correctly explained.
31-45	31-45	At least two of the three shocks correctly analysed with the fixed- S /adjusting- P -or-reserves logic, but the reasoning for at least one is incomplete.
16-30	16-30	Only one shock correctly analysed under the fixed-rate mechanics, or the answer incorrectly describes S itself changing in response to the shocks.
0-15	0-15	No credible fixed-exchange-rate analysis given.

Question A1 Total: 100 Marks

Section A · Question 2

PPP and the real exchange rate, and two cross-currency triangulations — 100 marks

A2(a)**[40 marks]**

Explain the purchasing power parity theory of exchange rate determination. What does this imply for the Real Exchange Rate?

A2(b)**[30 marks]**

Given the following interbank FX rates, what is the GBP/CHF rate? EUR/GBP 0.6642/45, EUR/CHF 1.5745/48.

A2(c)**[30 marks]**

Given the following interbank FX rates, what is the GBP/SGD rate? GBP/USD 1.6523/28, USD/SGD 1.7585/1.7600.

FULL MODEL ANSWER

Purchasing power parity theory: PPP holds that exchange rates adjust to equalise the purchasing power of different currencies — in its absolute form, $P = SP^*$, the price of an identical basket of goods should be the same everywhere once converted to a common currency via the exchange rate S . The theory rests on the **law of one price** generalised across an entire basket of goods: if a basket were cheaper in one country once converted to a common currency, arbitrageurs would buy there and sell where it's expensive, and this arbitrage activity — bidding up demand (and prices) where goods are cheap and increasing supply (pushing down prices) where they're expensive — would continue until the price gap closed.

The relationship: the real exchange rate is defined as $Q = SP^*/P$ — the nominal exchange rate S adjusted for relative price levels. Absolute PPP ($P = SP^*$) holds exactly if and only if **$Q = 1$** : by direct substitution, $Q = SP^*/P = SP^*/(SP^*) = 1$. So PPP and the real exchange rate are two ways of describing the same underlying claim — PPP asserts that the real exchange rate is constant and equal to 1, meaning a given basket of goods costs the same everywhere once converted to a common currency, leaving no genuine difference in purchasing power once the nominal exchange rate is accounted for.

Why deviations of Q from 1 matter economically: when $Q \neq 1$, the real exchange rate is capturing genuine differences in relative competitiveness — a country whose real exchange rate has appreciated (Q risen, in a convention where higher Q means the country's goods are relatively more expensive once converted to a common currency) has become less price-competitive internationally, regardless of what its nominal exchange rate has done, since nominal exchange-rate movements can be offset or reinforced by relative inflation differences.

Mark Allocation — Part (a) (40 marks)

Tier	Marks	What separates this tier
31-40	31-40	Correctly states absolute PPP ($P=SP^*$) with the law-of-one-price foundation and arbitrage mechanism explained AND correctly derives $Q=1$ from PPP AND explains the economic meaning of the real exchange rate.
21-30	21-30	Correct PPP statement and $Q=1$ derivation, but the arbitrage mechanism underlying PPP is not explained.
11-20	11-20	Vague statement of PPP without the formula or the real exchange rate

implication.

0-10

0-10

No credible answer given.

Setting up the triangulation: EUR is the common **base** currency in both quoted pairs (EUR/GBP and EUR/CHF), so the cross rate is built by dividing the two quotes so the EUR term cancels — using the same bid-with-offer, offer-with-bid matching as any other triangulation, to protect the dealer's margin on the synthetic transaction.

Quote	Calculation	Result
GBP/CHF bid	$1.5745 \div 0.6645$	2.3695
GBP/CHF offer	$1.5748 \div 0.6642$	2.3710

Why this direction of division: the dealer's **bid** for GBP/CHF (buying GBP, paying CHF) is synthesised by first selling the GBP received for EUR (receiving EUR at a rate corresponding to the EUR/GBP **offer**, the less favourable side for the dealer as the "seller" in this leg), then selling that EUR for CHF (at the EUR/CHF **bid**). This gives the formula GBP/CHF bid = (EUR/CHF bid) $\div$ (EUR/GBP offer); the offer follows the mirror-image logic.

Result: the no-arbitrage GBP/CHF quote is **2.3695 / 2.3710**.

Mark Allocation — Part (b) (30 marks)

Tier	Marks	What separates this tier
23-30	23-30	Correctly identifies EUR as the common base currency to cancel AND correctly applies bid-with-offer, offer-with-bid triangulation AND arrives at 2.3695 / 2.3710 (or a consistent own-figure result).
15-22	15-22	Correct triangulation method, but an arithmetic error in the final figures.
8-14	8-14	Incorrect method (e.g. bid-with-bid) but carried through consistently.
0-7	0-7	No coherent triangulation attempted.

Setting up the triangulation — a genuinely different structure from Part (b): here USD is the **quote** currency in GBP/USD but the **base** currency in USD/SGD — this is a direct chain (GBP $\rightarrow$ USD $\rightarrow$ SGD), not a "common base, two ways" cross like Part (b). A direct chain like this is built by **multiplying** the two rates (with matching bid-with-bid, offer-with-offer), not dividing.

Quote	Calculation	Result
GBP/SGD bid	1.6523×1.7585	2.9056
GBP/SGD offer	1.6528×1.7600	2.9089

Why bid pairs with bid here (not offer, unlike Part (b)): the dealer's **bid** for GBP/SGD (buying GBP, paying SGD) is synthesised by first selling the GBP received into USD — since the dealer is *selling* GBP into the interbank GBP/USD market, they receive the market's **bid** (1.6523 USD per GBP) — and then selling that USD into SGD, again receiving the market's **bid** (1.7585 SGD per USD, since the dealer is once more the seller in that leg too). Because both legs of this particular chain have the dealer on the *same side* (selling into each successive market), both use the bid; the offer calculation mirrors this with the dealer buying in both legs instead.

Result: the no-arbitrage GBP/SGD quote is **2.9056 / 2.9089**.

EXAM TECHNIQUE

Parts (b) and (c) are deliberately structured to require two different triangulation techniques — Part (b)'s EUR is a common *base* for both quoted pairs (divide, cross bid with offer), while Part (c)'s USD is quote in one pair and base in the other, forming a genuine chain (multiply, match bid with bid and offer with offer). Applying Part (b)'s division method to Part (c), or vice versa, produces a numerically different — and incorrect — answer, so identifying which structure you're facing before choosing a method is the first and most important step in both parts.

Mark Allocation — Part (c) (30 marks)

Tier	Marks	What separates this tier
23-30	23-30	Correctly identifies this as a direct chain (USD is quote in one pair, base in the other) requiring multiplication rather than division AND correctly matches bid-with-bid, offer-with-offer AND arrives at 2.9056 / 2.9089 (or a consistent own-figure result).
15-22	15-22	Correct multiplicative method, but an arithmetic error in the final figures, or bid/offer sides mismatched.
8-14	8-14	Incorrectly applies Part (b)'s division method to this chain structure, but carries the (wrong) method through consistently.
0-7	0-7	No coherent triangulation attempted.

Question A2 Total: 100 Marks

Section A · Question 3

The UIRP puzzle, and testing interest rate parity numerically — 100 marks

A3(a)

[40 marks]

Based on further readings, what is your understanding of the "UIRP Puzzle"?

A3(b)

[60 marks]

Does IRP hold in the below example? Show your workings. If IRP holds, is it said to be Covered or Uncovered? Why? 1st April 2023: UK 12-month interest rate 26%, US 12-month interest rate 5%; £100 available for deposit in both; Spot FX (£ price of \$) £0.50; Expected spot FX on 31st March 2024 (£ price of \$) £0.60.

EXAM TECHNIQUE

The "UIRP Puzzle" wasn't set out with its formal name in the uploaded course material, though the module's own quiz content directly discusses the underlying instability of uncovered interest parity. The explanation below follows the standard treatment (Fama, 1984) — also known as the "forward premium puzzle" or "forward premium anomaly" in the literature.

What the puzzle is: uncovered interest rate parity predicts that a currency with a higher interest rate should be expected to **depreciate** by approximately the size of the interest differential, so that the expected return to holding either currency, uncovered, is equalised. Empirically, this is not what happens on average: when researchers regress the actual future change in the exchange rate on the current interest differential (or equivalently the forward premium), the estimated slope coefficient is typically **close to zero or even significantly negative** — nowhere near the value of +1 that UIP predicts. In other words, high-interest-rate currencies tend, on average, to **appreciate** rather than depreciate as UIP would require, or at least depreciate by far less than the interest differential would predict.

Why this is a genuine puzzle, not just noise: the practical implication is that a simple **carry trade** strategy — borrowing in a low-interest-rate currency and investing, uncovered, in a high-interest-rate currency — has historically earned a positive average excess return over long samples, which directly contradicts UIP's prediction that such a strategy should earn zero expected excess return (any interest-rate advantage should be exactly offset, in expectation, by currency depreciation). This is a genuine, persistent, well-documented empirical regularity, not a one-off statistical fluke.

Proposed explanations in the literature

- **Time-varying risk premia:** UIP in its clean form assumes risk-neutral investors (see the EC3214 Summer 2025 scheme, Q. A1(c)) — if instead investors demand a risk premium for holding a currency, and that premium varies over time and happens to be correlated with the interest differential itself, the observed "UIP failure" may simply reflect the omitted, time-varying risk premium rather than any failure of rational behaviour.
- **The "peso problem":** if there is a small, but non-trivial, probability of a large currency devaluation that simply didn't materialise within a given sample period, realised returns in that sample will look biased relative to what was genuinely, rationally expected at the time — the small-sample average return overstates the true, full-distribution expected return once the low-probability, large-loss scenario is properly accounted for.
- **Limits to arbitrage:** exploiting the puzzle (via carry trades) requires taking on genuine risk and tying up capital; institutional constraints, funding costs, and periodic sharp carry-trade unwind episodes (where the strategy suffers large, sudden losses) may prevent the puzzle from being fully arbitrated away even if it represents a genuine pricing anomaly.

Mark Allocation — Part (a) (40 marks)

Tier	Marks	What separates this tier
31-40	31-40	Correctly states the UIP prediction and the empirical finding that contradicts it (high-rate currencies tend to appreciate, not depreciate, or depreciate far less than predicted) AND connects this to the carry trade's positive average excess return AND at least two specific proposed explanations (risk premia, peso problem, limits to arbitrage) correctly given.
21-30	21-30	Correctly states the empirical UIP failure and its carry-trade implication, with at least one explanation given.

11-20	11-20	Vague reference to "UIP doesn't always hold" without the specific empirical regularity (direction of the bias) or any explanation.
0-10	0-10	No credible answer given.

Setting up the two strategies: compare investing £100 in the UK for 12 months against converting to USD, investing in the US, and converting back at the **expected** future spot rate (note: an *expected* rate, not a contractually agreed forward rate — this distinction matters directly for the second part of the question).

Strategy	Calculation	Result
(A) Invest £100 in the UK at 26%	100×1.26	£126.00
Convert £100 to USD at spot (£0.50 per \$)	$100 \div 0.50$	\$200.00
Invest \$200 in the US at 5%	200×1.05	\$210.00
(B) Convert back at expected spot (£0.60 per \$)	210×0.60	£126.00

Does IRP hold? Yes — exactly. Both strategies produce precisely £126.00, with no gap at all between the UK strategy and the US-then-convert strategy. Using the exact multiplicative relationship — $(1+r_{UK}) = (1+r_{US}) \times (S_{\text{expected}}/S_{\text{spot}})$ — the two sides are precisely equal here: $1.26 = 1.05 \times (0.60/0.50) = 1.05 \times 1.20 = 1.26$.

EXAM TECHNIQUE

Worth noting explicitly: the commonly-used *linear approximation* of IRP ($r_{UK} - r_{US} \approx \text{expected \% change in the exchange rate}$) gives $26\% - 5\% = 21\%$, compared with an expected depreciation of $(0.60 - 0.50)/0.50 = 20\%$ — a 1 percentage point gap. This does **not** mean IRP fails here; it means the linear approximation is imprecise at these unusually large interest rates and expected currency movements (26% and a 20% expected depreciation are both far larger than the small values for which the linear approximation is usually a close fit). The exact, multiplicative relationship — the one that genuinely matters — holds precisely, as shown above.

Covered or Uncovered — and why: Uncovered. The calculation uses the *expected* future spot rate (£0.60 per \$, described explicitly as an "expected spot FX" figure), not a forward rate contracted today. Because this is a subjective expectation about an uncertain future outcome rather than a rate locked in by a forward contract, the position carries genuine **exchange rate risk** — the actual spot rate on 31 March 2024 could turn out to differ from what was expected on 1 April 2023, in which case the two strategies' realised returns would *not* match, even though they match exactly under the assumption that the expectation is realised precisely. This is exactly the distinction drawn in the EC3214 Summer 2025 scheme, Q. A1(c)-(d): covered interest parity uses a contracted, risk-free forward rate and holds by pure no-arbitrage logic regardless of risk preferences; uncovered interest parity uses an uncertain expected rate and — as this question's own Part (a) demonstrates — does not reliably hold in the real world, even though it holds exactly in this particular hypothetical numerical example.

Mark Allocation — Part (b) (60 marks)

Tier	Marks	What separates this tier
46-60	46-60	Both strategies correctly calculated, arriving at £126.00 for each AND correctly concludes IRP holds exactly AND correctly identifies this as Uncovered IRP, with the reasoning (expected rate vs. contracted forward rate; presence of genuine exchange rate risk) correctly explained.

31-45	31-45	Both strategies correctly calculated and IRP correctly concluded to hold, but the Covered/Uncovered identification is incorrect or unexplained.
16-30	16-30	Only one strategy correctly calculated, or an arithmetic error leads to an incorrect "IRP does not hold" conclusion.
0-15	0-15	No coherent calculation attempted.

Question A3 Total: 100 Marks